

Scalable and conformal nickel oxide for efficient perovskite solar modules

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Scalable, conformal nickel oxide (NiO_x) is essential for uniform charge extraction in large-area inverted perovskite modules, yet traditional deposition comes with a trade-off between structural coverage and surface reactivity. Here we report a coordination-regulated strategy based on chemical bath deposition that resolves this bottleneck by governing the NiO_x crystallization kinetics. By leveraging the high thermodynamic stability of Ni^{2+} –potassium tartrate (Pta) complexes, we suppress disorderly aggregation and halve the hydrolysis rate constant to $0.124 \text{ l mol}^{-1} \text{ min}^{-1}$. This precise kinetic control yields dense and conformal films with refined 4.8-nm nanocrystallites. This nanostructural refinement maximizes the surface density of hydroxyl groups, providing robust anchoring sites where self-assembled monolayers can form a chemically cohesive and defect-suppressed buried interface. Our Pta- NiO_x strategy delivers inverted devices with a champion power conversion efficiency of 27.40% (certified reverse-scan power conversion efficiency of 27.35%) and module efficiencies of 23.42% (17.7 cm^2) and 21.97% (81.5 cm^2). Small-area devices retained 90% of their initial power conversion efficiency after 2,656 h, and 17.7 cm^2 mini-modules retained over 90% after 1,200 h under the ISOS-L-II protocol.

Inverted (p–i–n) perovskite solar cells (PSCs) combine high efficiency with favourable operational stability^{1,2}. Nickel oxide (NiO_x) is widely used as a hole-selective interlayer in devices exceeding 27% efficiency because it can isolate the transparent electrode and provide a chemically robust surface for phosphonic acid self-assembled monolayers (SAMs), including [4-(3,6-dimethyl-9H-carbazol-9-yl)butyl] phosphonic acid (Me-4)³. On textured fluorine-doped tin oxide (FTO), a continuous NiO_x coating suppresses electron leakage, and the surface hydroxyl groups support dense SAM assembly and efficient hole extraction⁴.

Scaling this architecture requires NiO_x films that combine physical conformality with surface chemical reactivity^{5,6}. Spin-coating forms high-quality layers on small, flat substrates but becomes non-uniform over large textured areas, where fluid depletion at protrusions and coffee-ring flows create uncovered regions and shunt pathways^{7–9}. Magnetron sputtering has better large-area coverage, but energetic deposition often yields hydroxyl-deficient, chemically less active NiO_x with insufficient Ni^{3+} states for strong SAM binding^{10–13}. The resulting discontinuous or weakly coupled NiO_x /SAM interfaces are susceptible

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to SAM desorption, non-uniform charge extraction and operational degradation, limiting reliable perovskite scaling¹⁴.

Chemical bath deposition (CBD) offers the solution-based, topography-following growth of metal oxides^{15,16}. For NiO_x , however, the OH^- -induced hydrolysis of Ni^{2+} simultaneously drives substrate nucleation and uncontrolled precipitation in solution¹⁷. Rapid bulk nucleation favours cluster-by-cluster aggregation and coarse particles rather than the dense nanocrystalline framework required for an electronically selective buried interface¹⁸. Alkanolamine ligands can slow hydrolysis, but precise control of precursor activity and nucleation remains difficult^{19,20}.

Here we analysed the crystallization kinetics of CBD NiO_x and identified the hydrolysis rate of Ni^{2+} precursors as a key determinant of film conformality and surface chemistry. Screening ligands with different denticities identified that potassium tartrate (Pta) is an effective bidentate kinetic regulator. The high thermodynamic stability of Ni^{2+} -Pta complexes (cumulative stability constant $\log\beta = 3.866$) halved the hydrolysis rate constant k to $0.124 \text{ l mol}^{-1} \text{ min}^{-1}$, shifting deposition from uncontrolled precipitation to regulated crystallization. This control yielded uniform $17 \pm 2.2\text{-nm}$ films on textured substrates and refined 4.8-nm nanocrystallites, producing a hydroxyl-rich surface for dense SAM anchoring. The resulting Pta- NiO_x /SAM architecture delivered a champion power conversion efficiency (PCE) of 27.40% (certified reverse-scan PCE of 27.35%), and devices retained 90% of their initial performance after 2,656 h under the ISOS-L-II protocol. Modules achieved PCEs of 23.42% and 21.97% for active areas of 17.7 cm^2 and 81.5 cm^2 , respectively, with a T_{90} (the time required for the PCE to decline to 90% of its initial value) exceeding 1,200 h for the 17.7-cm^2 module.

Results and discussion

Ligand-regulated growth mechanism of conformal NiO_x films

In CBD of NiO_x , crystallization is governed by the hydrolysis equilibrium of Ni^{2+} precursors, which controls nucleation and growth through the formation of nickel hydroxide intermediates. In conventional CBD systems, abundant free Ni^{2+} reacts rapidly with OH^- , inducing bulk nucleation that outcompetes interface-directed growth. Effective kinetic regulation should, therefore, limit hydrolytically active Ni^{2+} rather than suppress OH^- . Guided by classical coordination chemistry, we introduced multidentate ligands that sequester and gradually release metal cations through chelation, thereby buffering Ni^{2+} hydrolysis and promoting regulated, interface-dominated nickel hydroxide growth (Fig. 1a and Supplementary Figs. 1 and 2). To test this hypothesis and establish a coordination hierarchy, we compared three ligands with distinct denticities: erythritol (Ery, non-chelating), potassium D-gluconate (Per, monodentate-like) and potassium tartrate (bidentate).

To quantify ligand binding and its thermodynamic effects, we combined density functional theory (DFT) calculations with potentiometric and spectroscopic titrations. The calculated binding energies were -6.07 eV , -6.58 eV and -8.53 eV for the Ery, Per and Pta complexes, respectively (Supplementary Fig. 3). The potentiometric titration yielded cumulative stability constants ($\log\beta$) of 1.600, 1.904 and 3.866, corresponding to a complexed Ni^{2+} fraction of 88.96% for Pta at pH 9 (Fig. 1b, Supplementary Fig. 4 and Supplementary Note 1). Ultraviolet–visible (UV–vis) and $^1\text{H}/^{13}\text{C}$ nuclear magnetic resonance titrations supported the formation of a rigid, multidentate Ni^{2+} -Pta complex (Supplementary Figs. 5–7). This thermodynamic buffering lowered free Ni^{2+} activity and suppressed supersaturation during deposition.

Time-resolved UV–vis spectroscopy was then used to monitor $\text{Ni}(\text{OH})_2$ nucleation through the absorption feature near 200 nm (Fig. 1c)^{21,22}. Ery showed a rapid increase in absorbance and visible colloidal scattering, whereas Pta maintained a weak signal (Supplementary Fig. 8). The corresponding second-order rate constants k were $0.256 \text{ l mol}^{-1} \text{ min}^{-1}$ and $0.124 \text{ l mol}^{-1} \text{ min}^{-1}$ for Ery and Pta, respectively (Supplementary Figs. 9 and 10), showing that strong

chelation approximately halves the hydrolysis rate and suppresses burst precipitation.

Cross-sectional high-resolution transmission electron microscopy (HRTEM) showed that Pta-regulated hydrolysis produced a conformal, compact film, unlike the coarse aggregates observed in the Ery and Per samples (Fig. 1d and Supplementary Fig. 11). The average thicknesses of Ery-, Per- and Pta- NiO_x were $22 \pm 15.3 \text{ nm}$, $18 \pm 11.2 \text{ nm}$ and $17 \pm 2.2 \text{ nm}$, respectively, indicating improved film uniformity and surface coverage (Supplementary Fig. 12 and Supplementary Table 1). A crystallite-size analysis showed that the average crystallite size decreased from 19.2 nm for Ery- NiO_x to 4.8 nm for Pta- NiO_x (Fig. 1e,f), indicating that kinetic inhibition promotes high nucleation density and fine-grained growth, thereby increasing the specific surface area (Supplementary Fig. 13).

Because SAM anchoring depends on surface active sites, we examined how crystallite refinement alters the NiO_x surface chemistry. X-ray photoelectron spectroscopy (XPS) revealed progressive enrichment of Ni^{3+} , with the $\text{Ni}^{3+}/\text{Ni}^{2+}$ ratio reaching 3.33 for Pta- NiO_x (Supplementary Fig. 14 and Supplementary Table 2). We attribute this activation to the increased surface-to-volume ratio of the nanocrystallites, which promotes oxidation at grain boundaries²³. Consistently, an O 1s analysis showed that Pta- NiO_x had the highest hydroxyl oxygen (O_{OH}) content of 33.7% (Fig. 1g,h and Supplementary Table 3), such that it provides abundant sites for phosphonic acid condensation⁴. Ultraviolet photoelectron spectroscopy (UPS) showed that the valence band maximum shifted from -5.08 eV for Ery- NiO_x to -5.19 eV for Pta- NiO_x , consistent with enhanced O 2p and Ni 3d orbital hybridization. After Me-4 functionalization, the valence band maximum shifted to -5.14 eV and -5.39 eV , respectively, reducing the hole-extraction barrier E_b from 0.28 eV to 0.03 eV and facilitating interfacial charge transport (Supplementary Figs. 15 and 16). To benchmark the strategy, we compared Pta- NiO_x with conventional spin-coated and sputtered NiO_x . Spin-coated NiO_x showed comparable hydroxyl levels, whereas sputtered NiO_x was chemically less active and exhibited weaker SAM anchoring in solvent-washing tests (Supplementary Figs. 17 and 18 and Supplementary Table 4). Coupled with conformal coverage, the enhanced surface chemistry of Pta- NiO_x improved scaling robustness and reproducibility in 17.7 cm^2 mini-modules relative to both controls (Supplementary Figs. 19–21 and Supplementary Table 5).

Coordination-enhanced interfacial chemistry and SAM anchoring

We incorporated ligand-processed NiO_x films into $\text{NiO}_x/\text{Me-4}$ heterostructures and examined their morphology and SAM anchoring. Consistent with the crystallite refinement observed by TEM, scanning electron microscopy showed that Pta- $\text{NiO}_x/\text{Me-4}$ formed a continuous, conformal film without the agglomerates and voids present in Ery- $\text{NiO}_x/\text{Me-4}$ and Per- $\text{NiO}_x/\text{Me-4}$ (Supplementary Fig. 22). Atomic force microscopy (AFM) revealed a root-mean-square roughness of 27.4 nm for Pta- $\text{NiO}_x/\text{Me-4}$, substantially lower than that of the Ery-based control. Moreover, multi-point mapping across $5 \times 5 \text{ cm}^2$ substrates confirmed the macroscale uniformity (Supplementary Figs. 23 and 24). We attribute this planarization to regulated crystallization, which suppressed the stochastic $\text{Ni}(\text{OH})_2$ aggregation and smoothed sharp FTO protrusions, providing a uniform template for dense SAM anchoring.

We next examined whether the hydroxyl-rich Pta- NiO_x surface enhances the SAM anchoring density. P 2p signals confirmed molecular grafting on all substrates (Fig. 2b–d), whereas suppression of the Sn 4s signal at 138.9 eV for Pta- NiO_x indicated reduced FTO exposure and a more continuous interface. XPS mapping of phosphorus in Me-4 showed denser adsorption on Pta- NiO_x (Supplementary Fig. 25). AFM with infrared spectroscopy showed that the uniform hydroxyl distribution promoted homogeneous, high-density SAM anchoring (Supplementary Fig. 26). DFT calculations showed that the binding energy became more negative, from -0.78 eV for Ery- NiO_x to -4.26 eV for Pta- NiO_x , and the reaction pathway supported

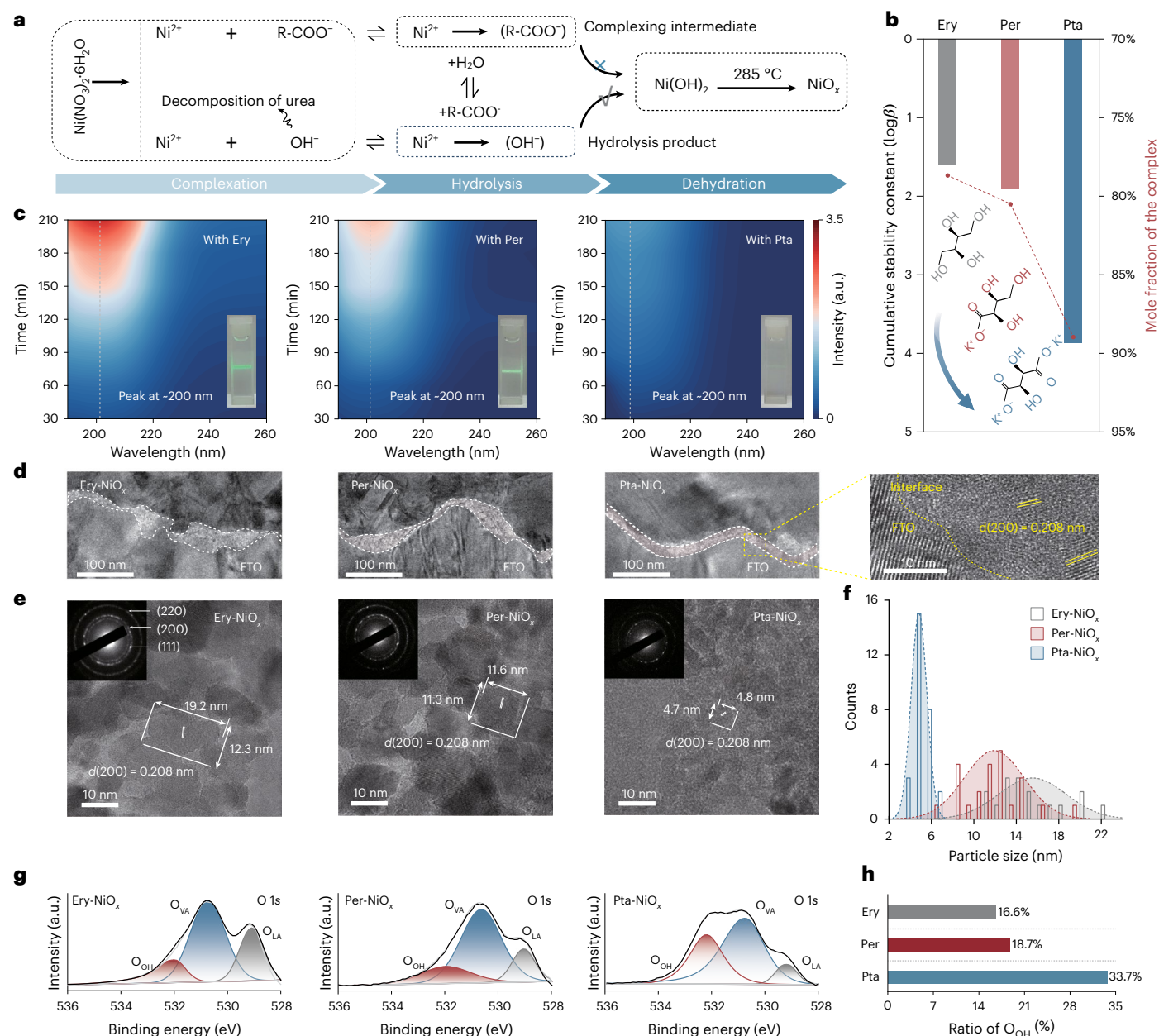


Fig. 1 | Coordination-regulated crystallization kinetics of NiO_x films.

a, Schematic illustration of the CBD growth mechanism, contrasting the ligand-mediated Ni^{2+} release and regulated crystallization (Pta system) with uncontrolled precipitation (control systems). **b**, Thermodynamic landscape of Ni^{2+} -ligand interactions, showing the cumulative stability constant ($\log\beta$, left axis) and species mole fractions (right axis) at pH 9. **c**, Time-resolved UV-vis absorption spectra monitoring the hydrolysis kinetics of Ery-, Per- and Pta-based precursors. Insets: Tyndall effect observations at 210 min (532-nm laser) reflecting the divergent nucleation rates. **d**, Cross-sectional HRTEM images of

NiO_x films on FTO substrates. **e**, HRTEM characterization of NiO_x nanoparticles and lattice fringes. Insets: corresponding selected-area electron diffraction (SAED) patterns of Ery- NiO_x , Per- NiO_x and Pta- NiO_x , respectively, with the diffraction rings indexed to the (111), (200) and (220) planes of NiO_x . $d(200)$, the interplanar spacing of the (200) lattice planes. **f**, Statistical crystallite size distributions derived from TEM analysis. **g**, XPS O 1s spectra of Ery- NiO_x , Per- NiO_x and Pta- NiO_x films, deconvoluted into lattice oxygen (O_{LA}), vacancy oxygen (O_{VA}) and hydroxyl oxygen (O_{OH}) species. **h**, Comparison of O_{OH} ratios in NiO_x films.

multidentate anchoring through sequentially deeper energy minima (Supplementary Figs. 27 and 28). Stronger anchoring also generated a larger interfacial dipole, favouring hole extraction. Molecular dynamics simulations indicated improved SAM coverage uniformity, with the phosphorus-atom distribution increasing from 61.35% to 64.05% (Supplementary Fig. 29). Cyclic voltammetry using a redox probe yielded a grafting density of 2.35×10^{13} molecules per square centimetre for Pta- NiO_x , 82.17% higher than that of the Ery control (Supplementary Fig. 30 and Supplementary Note 2). Multi-point contact-angle mapping across 5×5 cm² substrates confirmed that the

most uniform SAM coverage was on Pta- NiO_x (Supplementary Fig. 31). Kelvin probe force microscopy (KPFM) showed that FTO/Pta- NiO_x /Me-4 had the lowest mean contact potential difference of 331 mV, consistent with effective phosphonate- NiO_x bonding and uniform SAM anchoring (Fig. 2e-g). Together, these results identify the dense, hydroxyl-rich NiO_x layer as a prerequisite for high-density SAM assembly.

Interfacial resistance to solvent erosion and long-term ageing is critical for large-area devices. We compared XPS-derived coverage factors before and after vigorous rinsing with dimethylformamide (Supplementary Figs. 32-34, Supplementary Table 6 and

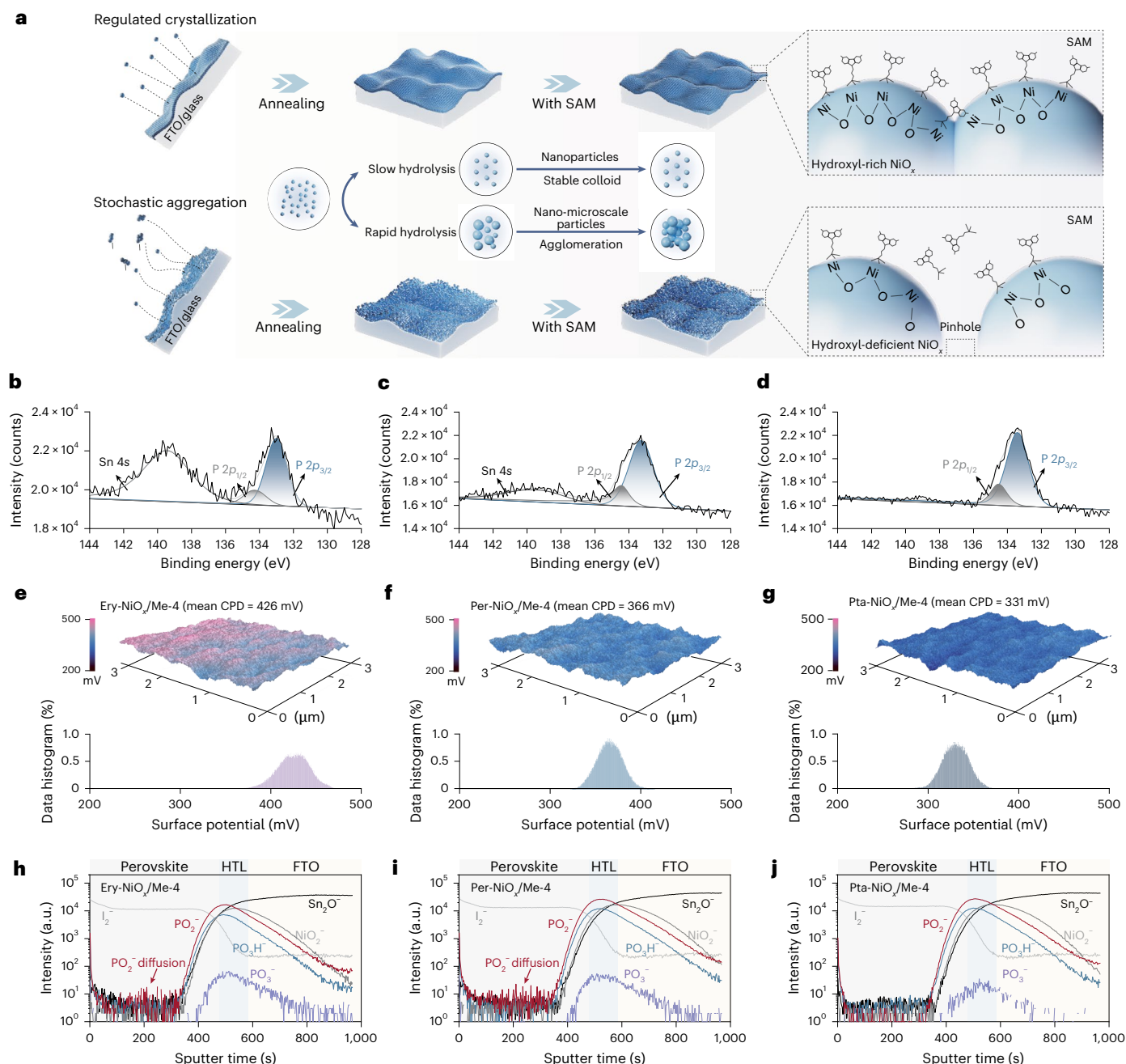


Fig. 2 | Interfacial characterization and stability of SAM-modified NiO_x films. **a**, Schematic illustration of the deposition pathways, highlighting the difference between stochastic aggregation (Ery- NiO_x and Per- NiO_x) and regulated growth (Pta- NiO_x), leading to enhanced SAM anchoring. **b–d**, High-resolution XPS P 2p core-level spectra indicating molecular grafting density for Ery- NiO_x /Me-4 (**b**), Per- NiO_x /Me-4 (**c**) and Pta- NiO_x /Me-4 (**d**). **e–g**, KPFM surface potential images (top) and corresponding contact potential difference (CPD) distributions

(bottom) of Ery- NiO_x (**e**), Per- NiO_x (**f**) and Pta- NiO_x (**g**) films modified with Me-4. **h–j**, Depth profiles of thermally aged Ery- NiO_x /Me-4/perovskite (**h**), Per- NiO_x /Me-4/perovskite (**i**) and Pta- NiO_x /Me-4/perovskite (**j**) stacks from time-of-flight secondary-ion mass spectrometry, tracking the spatial confinement of phosphonic acid fragments ($-\text{PO}_2^-$, $-\text{PO}_3^-$ and $-\text{PO}_3\text{H}^-$) to verify interfacial stability against diffusion. HTL, hole-transport layer.

Supplementary Note 3). Although all films showed some coverage loss, Pta- NiO_x /Me-4 retained a coverage factor of 8.37×10^{-2} , compared with 7.30×10^{-2} for Per- NiO_x /Me-4 and 8.35×10^{-3} for Ery- NiO_x /Me-4, indicating more robust Me-4 anchoring on the hydroxyl-rich Pta- NiO_x surface. We assessed SAM stability by depth profiling with time-of-flight secondary-ion mass spectrometry (Fig. 2h–j and Supplementary Note 4). In aged Ery- and Per-based samples, phosphonate fragments, including $-\text{PO}_2^-$, extended into the perovskite bulk, indicating SAM desorption and diffusion. By contrast, Pta-based samples aged at 85 °C for 240 h retained sharply localized phosphonate-fragment

signals at the interface. These results show that Pta- NiO_x stabilizes the SAM layer, suppressing interfacial diffusion and preserving long-term structural integrity.

We rationalize these improvements using a coordination-chemistry-based mechanistic model (Fig. 2a and Supplementary Fig. 35). Thermodynamic buffering of the precursor by Pta suppresses uncontrolled hydrolysis in conventional CBD, shifting growth from stochastic, colloid-dominated aggregation to regulated crystallization. This transition produces a compact, pinhole-free scaffold that suppresses leakage pathways, while crystallite refinement increases

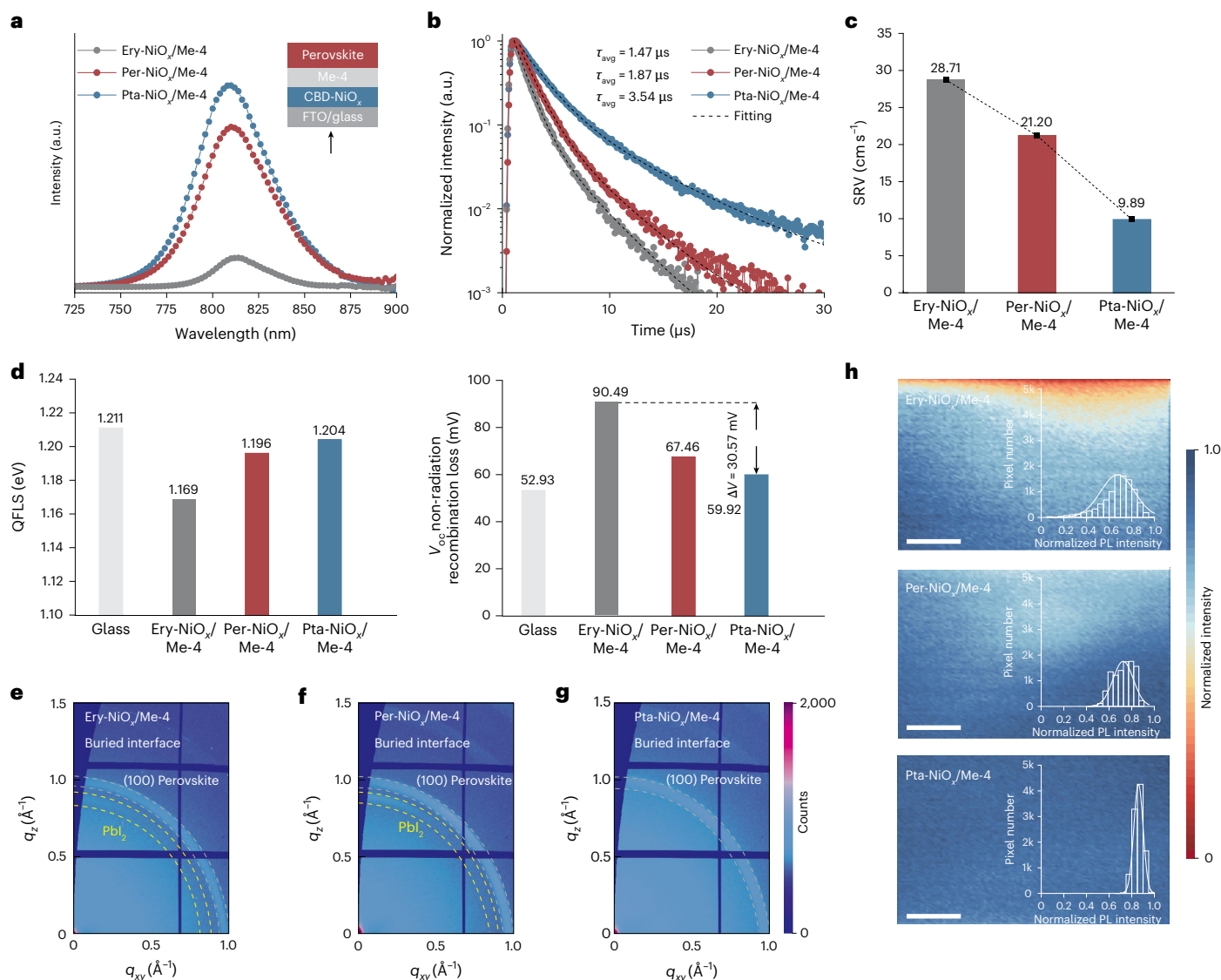


Fig. 3 | Interfacial charge-carrier dynamics and structural stability of NiO_x/Me-4 stacks. **a, b**, Steady-state PL spectra (**a**) and time-resolved PL decay curves (**b**) of perovskite films deposited on Ery-, Per- and Pta-NiO_x/Me-4 substrates. **c**, Statistical comparison of the surface recombination velocities (SRVs). **d**, Quasi-Fermi level splitting (QFLS) and V_{oc} losses derived from measurements of the PL quantum yield for perovskite films on various substrates. **e–g**, GIWAXS patterns of aged perovskite films on Ery-NiO_x/Me-4 (**e**), Per-NiO_x/Me-4 (**f**)

and Pta-NiO_x/Me-4 (**g**) after thermal ageing at 65 °C for 240 h under N₂. q_z and q_{xy} denote the out-of-plane and in-plane components of the scattering vector, respectively. **h**, PL mapping of the buried interfaces of perovskite films on Ery-, Per- and Pta-NiO_x/Me-4 substrates after thermal ageing at 65 °C for 240 h under N₂. Insets: the distributions of normalized PL intensity across the mapped pixels. Scale bars: 10 μm.

the surface-to-volume ratio and promotes surface oxidation, enriching reactive O_{OH} species. The resulting hydroxyl-rich, structurally coherent surface enables high-density SAM anchoring and forms a chemically robust, electronically selective interface.

Interfacial carrier recombination and perovskite film quality

To assess how the optimized NiO_x/SAM interface affects the charge-carrier dynamics, we examined recombination in the perovskite films. The highest emission intensity of the steady-state photoluminescence (PL) occurred for the film on Pta-NiO_x/Me-4, indicating suppressed interfacial non-radiative recombination, whereas quenched PL in the Ery- and Per-based controls indicated parasitic carrier loss at shunt sites arising from film discontinuities (Fig. 3a). Time-resolved PL showed that the carrier lifetime τ_{avg} increased from 1.47 μs for Ery to 3.54 μs for Pta, and the calculated surface recombination velocity decreased from 28.71 cm s⁻¹ to 9.89 cm s⁻¹ (Fig. 3b,c, Supplementary Table 7 and Supplementary Notes 5 and 6). Measurements of the PL quantum yield

gave a peak quasi-Fermi level splitting of 1.204 eV for Pta-NiO_x/Me-4 and a non-radiative open-circuit voltage V_{oc} loss of 59.92 mV, compared with 90.49 mV for the Ery control (Fig. 3d and Supplementary Figs. 36 and 37). Similar optical absorption profiles indicate that the V_{oc} gain arises from electronic rather than optical effects, and 25-point mapping confirmed the uniform transmittance of $81.19 \pm 0.35\%$ across $5 \times 5 \text{ cm}^2$ (Supplementary Figs. 38 and 39). Space-charge-limited current measurements on hole-only devices showed that Pta-NiO_x/Me-4 had the lowest trap-filled limit voltage $V_{TFL} = 0.1637 \text{ V}$, corresponding to a trap density of $8.36 \times 10^{14} \text{ cm}^{-3}$ (Supplementary Fig. 40 and Supplementary Note 7), consistent with a high-quality, defect-mitigated heterojunction²⁴.

Beyond defect mitigation, the optimized Pta-NiO_x/Me-4 interface increased the built-in potential V_{bi} to 1.04 V. Cross-sectional KPFM showed no interfacial potential spike and a potential drop of 164.1 mV (Supplementary Figs. 41 and 42). Femtosecond transient absorption spectroscopy showed suppressed non-radiative recombination and accelerated hole transfer, consistent with simulations

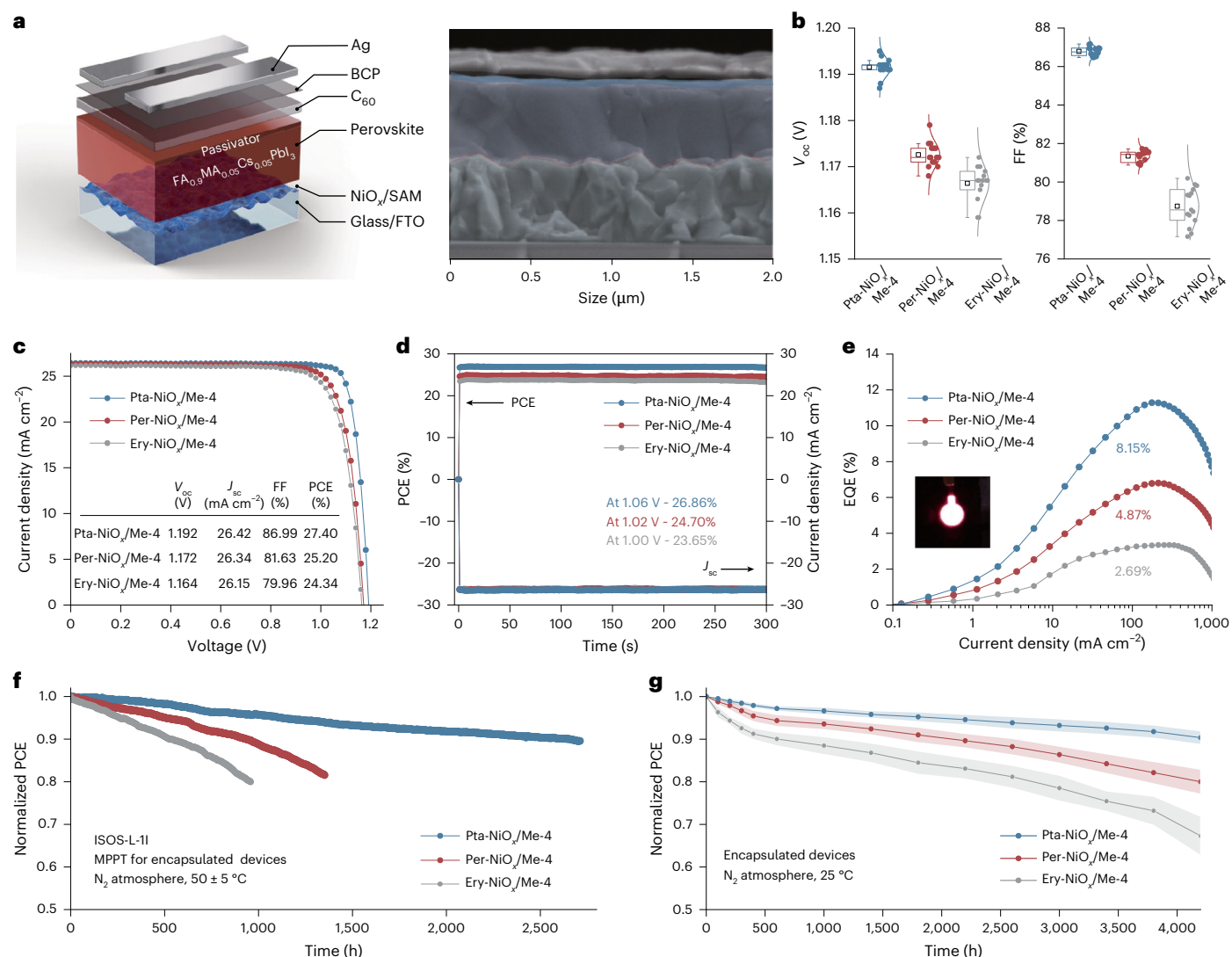


Fig. 4 | Photovoltaic performance of PSCs. **a**, Schematic of the p-i-n device architecture and a representative cross-sectional scanning electron microscopy image of a fabricated PSC. **b**, Statistical distributions of V_{oc} and FF for PSCs incorporating Ery-, Per- and Pta-NiO_x/Me-4 films. The box plots show the median (centre line), 25th to 75th percentiles (box bounds), minima and maxima (whiskers) and individual data points (circles). Data from $n = 15$ independent devices per group are shown. **c**, $J-V$ curves of the champion devices based on each NiO_x/Me-4 configuration. Inset: the corresponding V_{oc} , J_{sc} , FF and PCE values. **d**, Stabilized power output and current density tracking for each NiO_x/Me-4

device over 300 s. **e**, EQE_{EL} measurements for the corresponding devices. Inset: photograph of the electroluminescence from a representative device under electrical injection. **f**, MPPT performance under continuous illumination in an N₂ environment. **g**, Long-term stability of encapsulated devices stored in the dark under N₂ at 25 °C. The dark lines represent the mean normalized PCE, and the shaded regions represent ± s.d. Data were obtained from $n = 8$ independently fabricated devices per group, all measured in the same testing batch.

BCP, bathocuproine; FA, formamidinium; MA, methylammonium.

using the Solar Cell Capacitance Simulator (SCAPS-1D) to model the enhanced built-in field (Supplementary Figs. 43–45). Transient photovoltage measurements showed that the carrier lifetime increased from 0.03 μ s for Ery-NiO_x/Me-4 to 1.86 μ s for Pta-NiO_x/Me-4 (Supplementary Fig. 46). Dark current density versus voltage ($J-V$) measurements showed reduced shunt leakage, suppressed trap-assisted recombination and improved diode behaviour (Supplementary Fig. 47, Supplementary Table 10 and Supplementary Note 8). Electrochemical impedance spectroscopy found that the highest recombination resistance ($R_{rec} = 5.51$ k Ω) and lowest series resistance ($R_s = 18.3$ Ω) were for the Pta system (Supplementary Fig. 48 and Supplementary Table 11). Together, these results establish that Pta-NiO_x/Me-4 is an efficient hole-selective contact with reduced parasitic losses.

Suppressed interfacial recombination is associated with improved long-term stability. Grazing-incidence wide-angle X-ray scattering

(GIWAXS) at an incidence angle of 0.2° showed weakened perovskite (100) peaks and emerging PbI₂ signatures in aged Ery-NiO_x/Me-4 and Per-NiO_x/Me-4 samples, indicating localized phase decomposition (Fig. 3e–g and Supplementary Fig. 49). An X-ray diffraction analysis corroborated these results (Supplementary Fig. 50). We attribute these structural changes to surface inhomogeneity and incomplete SAM coverage, which permit detrimental perovskite/metal oxide interactions. By contrast, Pta-NiO_x/Me-4 showed no detectable PbI₂ after 720 h of ambient ageing (Supplementary Fig. 51). PL imaging from the buried-interface side showed reduced, spatially non-uniform emission in the Ery- and Per-based films, consistent with increased interfacial defect densities, whereas Pta-NiO_x/Me-4 showed no appreciable PL quenching or inhomogeneity²⁵ (Fig. 3h). This stability is attributed to the conformal, hydroxyl-rich NiO_x surface, which supports a robust, compact SAM. N 1s XPS after thermal ageing further showed that Pta-NiO_x/Me-4 most effectively suppressed formamidinium

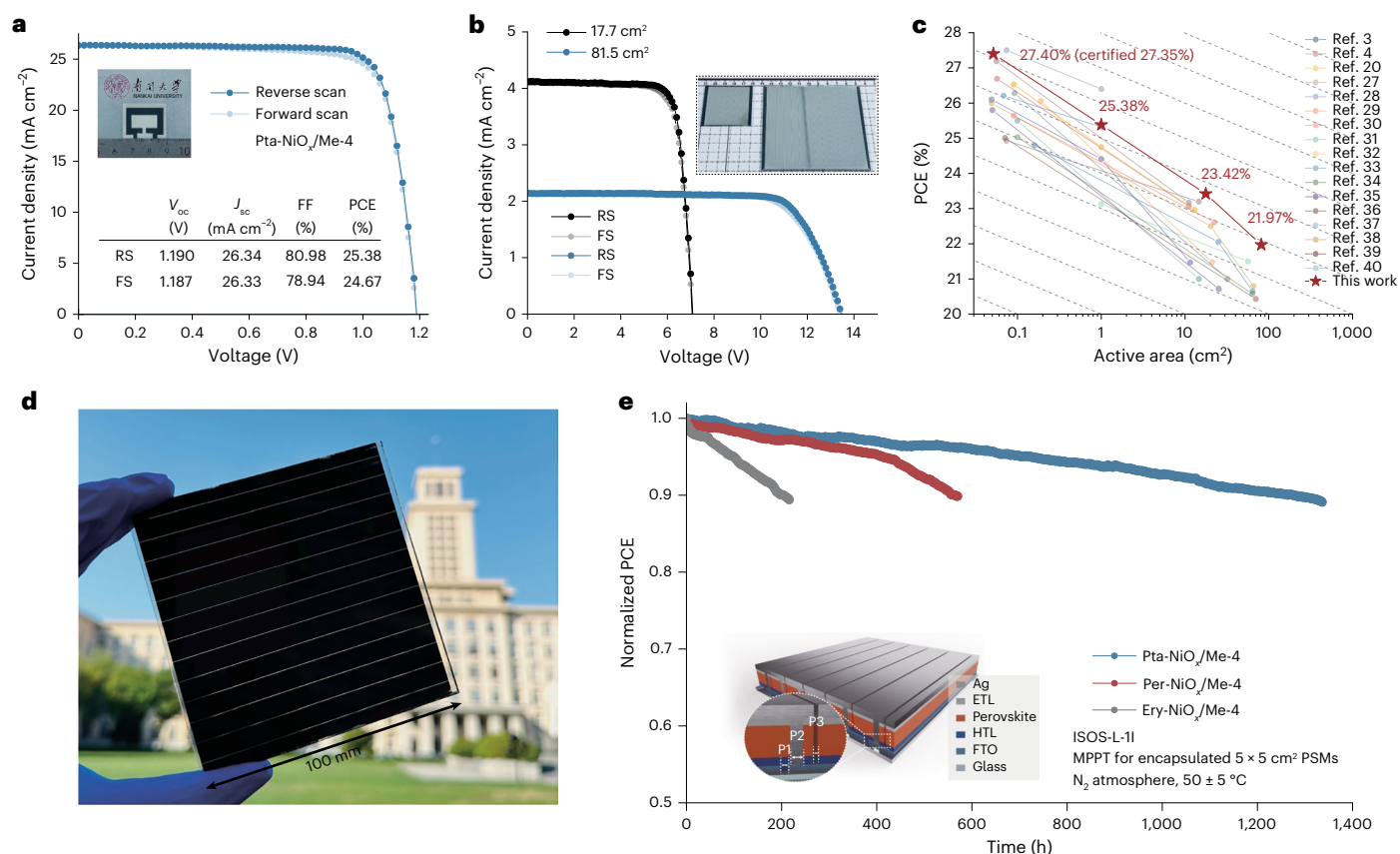


Fig. 5 | Scalable photovoltaic performance of cells and modules. a, J - V curves of Pta- $\text{NiO}_x/\text{Me-4}$ devices with an active area of 1 cm^2 . Inset: representative photograph of the 1 cm^2 device. **b,** J - V curves of Pta- $\text{NiO}_x/\text{Me-4}$ mini-modules with active areas of 17.7 cm^2 and 81.5 cm^2 . **c,** Comparison of PCEs for PSCs and perovskite solar modules (PSMs) with different active areas. **d,** Digital

photograph of the slot-die-printed $10 \times 10 \text{ cm}^2$ mini-module. **e,** Long-term MPPT operational stability of encapsulated mini-modules under simulated 1-sun illumination in an N_2 environment. Inset: schematic illustration of the module architecture and P1-P3 laser-scribed interconnections. FS, forward scan; RS, reverse scan; ETL electron-transport layer.

deprotonation, thereby limiting direct-contact-induced degradation and buried-interface void formation (Supplementary Figs. S2 and S3).

Device performance and stability

We fabricated p-i-n PSCs with the architecture FTO/ $\text{NiO}_x/\text{Me-4}$ /perovskite/passivator/ C_{60} /bathocuproine/Ag (Fig. 4a). The CBD conditions ($\text{Ni}^{2+}:\text{Pta} = 4:1$ at 70°C) were optimized for film quality and reproducibility (Supplementary Fig. S4). The conformal NiO_x morphology and dense SAM anchoring increased the V_{oc} , fill factor (FF) and short-circuit current density J_{sc} relative to the Ery and Per controls (Fig. 4b and Supplementary Fig. S5). The champion Pta- $\text{NiO}_x/\text{Me-4}$ device achieved a PCE of 27.40% ($V_{oc} = 1.192 \text{ V}$, $J_{sc} = 26.42 \text{ mA cm}^{-2}$ and $\text{FF} = 86.99\%$) with negligible hysteresis and a stabilized power output of 26.86% over 300 s under AM 1.5G illumination (Fig. 4c,d and Supplementary Fig. S6). The external quantum efficiency (EQE)-integrated J_{sc} agreed with the J_{sc} obtained from J - V measurements, and the spectrum indicated a bandgap of 1.53 eV (Supplementary Fig. S7). Independent certification yielded a PCE of 27.35%, comparable with the efficiencies of state-of-the-art inverted PSCs (Supplementary Fig. S8 and Supplementary Table 12).

Electroluminescence measurements showed an EQE_{EL} of 8.15% for the Pta-based device, compared with 4.87% and 2.69% for the Per- and Ery-based devices, respectively²⁶ (Fig. 4e and Supplementary Fig. S9). This enhancement corresponded to a 28.6-mV reduction in non-radiative V_{oc} loss (Supplementary Fig. S60 and Supplementary Note 9). Light-intensity-dependent V_{oc} measurements yielded an ideality factor $n = 1.167$ for the Pta-based device, which was lower than those of the Per-based (1.502) and Ery-based (1.665) devices

(Supplementary Fig. S61), indicating suppressed trap-assisted recombination consistent with the high FF of 86.99% (Supplementary Fig. S62 and Supplementary Note 10).

Under continuous 1-sun illumination during maximum power point tracking (MPPT) following the ISOS-L-II protocol at $50 \pm 5^\circ\text{C}$, Pta- $\text{NiO}_x/\text{Me-4}$ devices retained 90% of their initial PCE (26.82%) after 2,656 h (Fig. 4f and Supplementary Table 13), whereas Per- $\text{NiO}_x/\text{Me-4}$ and Ery- $\text{NiO}_x/\text{Me-4}$ devices retained 90.42% and 81.63%, respectively, after 900 h. After thermal ageing at 65°C for 1,000 h, conductive atomic force microscopy (c-AFM) found preserved conductive networks with a current of 43.5 nA, whereas KPFM showed only a 25-mV shift in the contact potential difference (Supplementary Figs. S63 and S64). Under N_2 storage, Pta- $\text{NiO}_x/\text{Me-4}$ devices retained 90% of their initial average PCE (27.26%) for over 4,200 h (Fig. 4g).

Scalable large-area solar cells

The simplicity and process compatibility of CBD support the scalable fabrication of Pta- $\text{NiO}_x/\text{Me-4}$. Devices with a 1 cm^2 active area achieved a champion PCE of 25.38% and a stabilized power output of 25.14%, with an average PCE of 25.27% across 15 independent cells (Fig. 5a and Supplementary Figs. S65 and S66).

We fabricated mini-modules on $5 \times 5 \text{ cm}^2$ and $10 \times 10 \text{ cm}^2$ substrates using scalable solution processing, including CBD and slot-die coating (Fig. 5d). Optimized P1, P2 and P3 laser-scribing patterns yielded a geometric fill factor (GFF) of 97.8%. Scanning electron microscopy and energy-dispersive X-ray spectroscopy mapping confirmed the well-defined scribing boundaries without elemental cross-diffusion or delamination (Supplementary Figs. S67 and S68).

The homogenized NiO_x/SAM interface mitigated printing-induced defects, enabling mini-module efficiencies of 23.42% and 21.97% on substrates with active areas of 17.7 cm² and 81.5 cm², respectively (Fig. 5b and Supplementary Table 14). Nine-region measurements showed uniform conductivity across the 5 × 5 cm² films, and 12 independent module batches yielded an average efficiency of 23.08 ± 0.35%. Uniform PL intensity and *J*–*V* characteristics were also observed across the 10 × 10 cm² substrates (Supplementary Figs. 69–72). These results represent state-of-the-art efficiencies for small-area cells and perovskite solar modules^{3,4,20,27–40} (Fig. 5c).

The Pta-NiO_x system retained its operational robustness at the module level. Under continuous 1-sun illumination during MPPT, the encapsulated 17.7 cm² mini-module retained over 90% of its initial PCE after 1,200 h, outperforming the Ery- and Per-based modules and showing competitive stability among reported perovskite solar modules (Fig. 5e and Supplementary Table 15). Pta-NiO_x/Me-4 modules also retained 93.2% of their initial efficiency after 500 h at 65 ± 5 °C (Supplementary Fig. 73). Under a reverse bias, the uniform Pta-NiO_x interface suppressed localized weak spots, delayed avalanche breakdown and maintained negligible dark leakage current (Supplementary Fig. 74). This interfacial uniformity also improved tolerance to partial-shading-induced reverse bias. Pta-NiO_x/Me-4 modules retained 80% of their initial PCE after 500 h, whereas the Per- and Ery-NiO_x/Me-4 modules degraded within 100 h (Supplementary Fig. 75).

Conclusion

Transitioning from stochastic aggregation to regulated crystallization is essential for conformal NiO_x growth on textured substrates. Thermodynamic buffering through Ni²⁺-Pta coordination ($\log\beta = 3.866$) halves the hydrolysis rate, overcoming the trade-off between structural conformality and surface reactivity. The resulting dense, hydroxyl-rich NiO_x promotes high-density SAM anchoring and forms a cohesive, defect-suppressed buried interface. This strategy enables a PCE of 27.40% (certified reverse-scan PCE of 27.35%) and module efficiencies of 23.42% and 21.97% for active areas of 17.7 cm² and 81.5 cm², respectively, together with long-term stability under ISOS protocols. This precise control of the crystallization kinetics therefore provides a viable route to scalable, stable perovskite photovoltaic modules.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41566-026-02012-z>.

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Methods

Materials

Caesium iodide (CsI, 99.9%), methylammonium iodide (99.99%), formamidinium iodide (99.99%), 3-(methylthio)propylamine hydroiodide (3MTPAI, 98%) and propane-1,3-diammonium iodide (PDAl₂, 98%) were purchased from Great Cell Solar Materials. Methylammonium chloride (>99.5%) was purchased from Luminescence Technology. Lead iodide (PbI₂, 99.999%) was purchased from Advanced Election Technology. Me-4 (99%) were purchased from TCI. Potassium hydroxide (KOH, 99.99%), nickel (II) nitrate hexahydrate (Ni(NO₃)₂·6H₂O, 99.99%), Pta (99%) and urea (>99%) were purchased from Aladdin. Per (98%) and Ery (99%) were purchased from Leyan. Bathocuproine (98.0%) and C₆₀ (99.5%) were purchased from Sigma-Aldrich. All reagents were used without further purification.

Small-area device fabrication

Hole-transport layer. FTO substrates were cleaned and treated with UV-ozone for 10 min. The three NiO_x precursor solutions were prepared separately by dissolving Ni(NO₃)₂·6H₂O (726.8 mg) with Ery (76.0 mg), Per (108.8 mg) or Pta (152.8 mg) in deionized water. Urea (400 mg) was added, and each solution was diluted with deionized water to a final volume of 200 ml, followed by adjustment of the pH to 8.0–8.5. CBD was performed in a glass vessel with internal dimensions of 16 × 16 × 8 cm³ using a freshly prepared precursor solution for each batch. The substrates were mounted horizontally, with the deposition surfaces facing downwards, on format-specific open-frame polytetrafluoroethylene holders with identical outer dimensions of 14 × 14 cm². The substrates were supported only along their edges, leaving the deposition surfaces fully exposed to the precursor solution. Twenty-five 2 × 2 cm² substrates were arranged in a 5 × 5 array, four 5 × 5 cm² substrates were in a 2 × 2 configuration or one 10 × 10 cm² substrate was placed at the centre of the holder, maintaining a total substrate area of 100 cm² per batch. No stirring was applied, and the vessel was maintained in a temperature-controlled oven at 70 ± 1 °C for 3.5 h. After deposition, the films were ultrasonically cleaned, dried under nitrogen and calcined at 285 °C for 2 h. After calcination, the NiO_x-coated substrates were transferred to an N₂-filled glovebox. For the small-area and 1 cm² devices, Me-4 (1 mg ml⁻¹ in ethanol) was spin-coated at 4,000 rpm for 30 s. For the 5 × 5 cm² and 10 × 10 cm² substrates, the same Me-4 solution was deposited by slot-die coating using a head-to-substrate gap of 150 μm and a coating speed of 12 mm s⁻¹. The ink supply rates were 3 μl s⁻¹ and 6 μl s⁻¹ for coating widths of 5 cm and 10 cm, respectively. The substrates were subsequently annealed at 105 °C for 10 min.

Perovskite layer. The precursor was formulated by dissolving CsI (19.5 mg), methylammonium iodide (11.9 mg), formamidinium iodide (232.2 mg) and PbI₂ (729.5 mg, 5.5% excess) in 1 ml dimethylformamide:dimethyl sulfoxide (4:1 v/v with 15 mol% methylammonium chloride added as a processing additive). The solution was spin-coated onto the NiO_x/Me-4 substrate via a two-step programme (1,000 rpm for 10 s then 4,000 rpm for 40 s), with 160 μl of chlorobenzene dispensed at 8 s before the end of the second step. After annealing at 105 °C for 35 min, the films were passivated by spin-coating a PDAl₂:3MTPAI mixture (1:1 w/w, 2 mg ml⁻¹ in isopropyl alcohol) and heated at 105 °C for 5 min.

Electron-transport layer and electrodes. Finally, C₆₀ (25 nm), bathocuproine (6 nm) and Ag (100 nm) were sequentially deposited via thermal evaporation under high vacuum (<5 × 10⁻⁶ Torr). The active area was defined as 0.0515 cm² using a shadow mask.

Large-area device fabrication

To evaluate the scalability of the CBD-NiO_x template, 1 cm² devices were fabricated on 2.5 × 2.0 cm² substrates. The procedure followed the small-area protocol with optimized volumetric adjustments: 100 μl

of perovskite precursor and 250 μl of chlorobenzene antisolvent were used to ensure uniform coverage. Surface passivation was similarly scaled using 100 μl of the PDAl₂:3MTPAI solution. All subsequent deposition and evaporation parameters were identical to those used for the small-area devices.

Large-area mini-module fabrication

FTO cleaning and NiO_x bath deposition followed the protocols established for small-area devices. For module scaling, the 1.3 M FA_{0.90}MA_{0.05}Cs_{0.05}PbI₃ precursor (15% methylammonium chloride and 5.5% excess PbI₂) was reformulated in a dimethylformamide:*N*-methyl-2-pyrrolidone (DMF:NMP 6:1 v/v) co-solvent system. The solution was deposited onto 5 × 5 cm² or 10 × 10 cm² substrates via slot-die printing (Shanghai Datamaker Technology) in an N₂ glovebox using a 50-μm head-to-substrate gap, a 20 mm s⁻¹ printing speed and an 11 μl s⁻¹ ink supply rate. The resulting wet films were immediately transferred to a vacuum chamber (10 Pa) for 50 s at 17–18 °C to regulate intermediate-phase crystallization, followed by annealing at 105 °C for 35 min. The passivation layer was similarly printed using 4 mM PDAl₂/3MTPAI (100 μm gap, 4 mm s⁻¹ speed and 150 μl s⁻¹ supply rate) and annealed at 105 °C for 5 min. To achieve a high GFF of 97.8%, serial interconnections for the 6- and 12-cell modules were patterned using a picosecond laser (Wuhan Yuanlu Optoelectronic Technology) to define P1, P2 and P3 patterns with scribe widths of 25 μm, 61 μm and 37 μm, respectively (dead area 149 μm). Note that an anti-reflection film (Shangyang Technology) was applied to the glass side of the FTO substrate for all devices.

Characterization

Crystal structures and surface morphologies were examined via X-ray diffraction (D8 Advance, Bruker, 10° min⁻¹), field-emission scanning electron microscopy (JSM-7900F, JEOL), HRTEM (FEI Talos F200X G2), AFM (Dimension Icon, Bruker) and AFM with infrared spectroscopy (NanoIR-3, Bruker). Chemical states were analysed by XPS (Nexsa, ThermoFisher), UPS (ESCALAB 250Xi, Thermo Scientific) and nuclear magnetic resonance (Avance III 400 MHz). Optical properties were recorded using a PerkinElmer Lambda 750S (UV–vis), an Edinburgh Instruments FLS 980 (steady-state and time-resolved PL under 600-nm excitation) and a QuantaMaster 8000 (Horiba, PL quantum yield, 532 nm, integrating sphere). BET surface areas were measured on a Micromeritics ASAP 2460. Electrochemical measurements, including electrochemical impedance spectroscopy, cyclic voltammetry and Mott–Schottky plots, were made using a workstation (CHI760). Femtosecond transient absorption spectroscopy measurements were performed using a pump–probe system (Helios, Ultrafast Systems) with a Ti:sapphire laser (Legend Elite-1K-HE, 800 nm, 25 fs, 4 mJ per pulse at 1 kHz). GIWAXS data were collected at BSRF-1W1A, Beijing Synchrotron Radiation Facility. Device *J–V* characteristics were obtained using a source meter (2400, Keithley) under calibrated AM1.5G illumination. The irradiance was set to 100 mW cm⁻², calibrated via a crystalline silicon reference cell equipped with a KG5 filter window. The solar simulator spectrum was meticulously adjusted to maintain a spectral mismatch within 100 ± 3% for every 50-nm interval across the 300–1,200-nm range. Before characterization, all devices underwent light soaking under constant illumination until a stabilized steady-state output was reached. Scan ranges were from 1.2 V to –0.02 V (step 0.02 V and delay 20 ms) for small-area cells and from 7.2 V or 15.0 V to –0.1 V (step 0.1 V and delay 20 ms) for mini-modules, in both forward and reverse directions. EQE and electroluminescence spectra were collected using Enlitech PECT-600 and Ocean Optics QE65 Pro systems. *J–V* measurements were conducted at 25 °C inside a temperature-controlled glovebox. Operational stability was tracked via MPPT at 50 ± 5 °C under continuous white LED illumination. Device encapsulation was performed using a glass cover and UV epoxy (Three Bond). Devices were then cured for 3 min under a 365-nm UV-LED source. Measurements

were conducted under the full simulated spectrum without a UV-cutoff filter. The LED intensity was periodically recalibrated to account for spectral ageing and to ensure constant irradiance.

DFT calculations

Surface adsorption calculations. The binding energies for SAM anchoring on NiO_x surfaces were calculated using the projector-augmented-wave method, with a plane-wave cutoff energy of 400 eV (refs. 41–43). The generalized gradient approximation in the Perdew–Burke–Ernzerhof form was used for the exchange–correlation functional, with an electronic self-consistent convergence criterion of 1.0×10^{-5} eV (ref. 44). The DFT-D3 (BJ) method was employed to account for van der Waals interactions⁴⁵. Intermediate structures along the anchoring pathway were generated by linear interpolation, with the highest-energy structure used as an approximate transition-state configuration.

Molecular electronic structure calculations. All molecular and ionic electronic structures were determined using the ORCA quantum-chemistry suite. Calculations employed the B3LYP exchange–correlation functional with Grimme’s DFT-D4 dispersion correction⁴⁶. Solvent effects were modelled implicitly via the CPCM continuum solvation approach⁴⁷. Geometry optimizations were carried out until the total energy change fell below 1×10^{-4} eV and the maximum force on any atom was under 5×10^{-4} eV Å⁻¹.

Molecular dynamics simulations

Molecular dynamics simulations were performed using the LAMMPS package with the universal force field⁴⁸. Van der Waals interactions were described using the Lennard–Jones potential, whereas electrostatic interactions were treated using a truncated Coulomb potential with a cutoff radius of 12.0 Å. Atomic charges were updated using the QEq method every 100 steps⁴⁹. After initial charge equilibration and energy minimization, production simulations were conducted in the NVT ensemble using a Nosé–Hoover thermostat⁵⁰. The NiO_x surface was modelled as a 20×20 supercell with four atomic layers, with the bottom two layers fixed and a 100-Å vacuum layer introduced to avoid periodic image interactions. For each hydroxyl-content model, 12 SAM molecules were vertically anchored on the surface, and another 88 molecules were randomly deposited using Packmol⁵¹.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The main data supporting the findings of this study are available in the Article and the Supplementary Information. All other data that support the findings of this study are available from the corresponding authors upon request.

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Author contributions

Y.J. and M.Y. supervised the project. H.Y., H.L., T.A., C.C., X.-Y.W., Y.J. and M.Y. conceived the idea and designed the experiments. H.Y. and Y. Wang were involved in all the experimental work. H.Y. and Z. Du fabricated the large-area cells and conducted the PCE measurements. Y. Wang fabricated the small-area cells and drew the schematics. S.L., D.W., Z. Ding and Y.H. discussed and analysed the data. X.H. measured the EQE. Q.Z. performed the TEM characterizations. Y. Wu and L.C. carried out the DFT calculations and molecular dynamics simulations. Finally, H.Y., H.L., Y.J. and M.Y. wrote the article, with input from all authors.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41566-026-02012-z>.

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► Experimental design

Please check the following details are reported in the manuscript, and provide a brief description or explanation where applicable.

1. Dimensions

Area of the tested solar cells	☒ Yes ☐ No	0.0515 and 1 cm ² for solar cells, 17.7 and 81.5 cm ² for solar modules. Explain why this information is not reported/not relevant.
Method used to determine the device area	☒ Yes ☐ No	A metal mask is used to defined the aperture area of solar cells and solar modules. Explain why this information is not reported/not relevant.

2. Current-voltage characterization

Current density-voltage (J-V) plots in both forward and backward direction	☒ Yes ☐ No	Figure 4c and Supplementary Figure S6.
Voltage scan conditions	☒ Yes ☐ No	All conditions are mentioned in the Methods section. Explain why this information is not reported/not relevant.
Test environment	☒ Yes ☐ No	The solar cells were tested at room temperature in a N ₂ -filled glove box. Explain why this information is not reported/not relevant.
Protocol for preconditioning of the device before its characterization	☒ Yes ☐ No	Prior to characterization, all devices underwent light soaking under constant illumination until a stabilized steady-state output was reached Explain why this information is not reported/not relevant.
Stability of the J-V characteristic	☒ Yes ☐ No	The stabilized power output tracking at the maximum power point was performed (Fig. 4d) Explain why this information is not reported/not relevant.

3. Hysteresis or any other unusual behaviour

Description of the unusual behaviour observed during the characterization	☒ Yes ☐ No	Negligible hysteresis was observed. Explain why this information is not reported/not relevant.
Related experimental data	☒ Yes ☐ No	Reverse and forward J-V scan can be found in Supplementary Figure S6. Explain why this information is not reported/not relevant.

4. Efficiency

External quantum efficiency (EQE) or incident photons to current efficiency (IPCE)	☒ Yes ☐ No	EQE curves were provided in Supplementary Figure S7. Explain why this information is not reported/not relevant.
A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator	☒ Yes ☐ No	The integrated J _{sc} obtained from EQE is agree well with the J _{sc} determined from the J-V measurement within an error of 2%. Explain why this information is not reported/not relevant.

For tandem solar cells, the bias illumination and bias voltage used for each subcell

☐ Yes
☒ No

Provide a description of the measurement conditions.

There are no tandem solar cells in this work.

5. Calibration

Light source and reference cell or sensor used for the characterization

☒ Yes
☐ No

Device J-V characteristics were obtained using a Keithley 2400 source meter under calibrated AM 1.5G illumination. Irradiance was set to 100 mW/cm², calibrated via a crystalline silicon reference cell equipped with KG5 filter window.

Explain why this information is not reported/not relevant.

Confirmation that the reference cell was calibrated and certified

☒ Yes
☐ No

The reference cell is a 2×2 cm calibrated solar cell made of crystalline silicon equipped with KG5 filter window.

Explain why this information is not reported/not relevant.

Calculation of spectral mismatch between the reference cell and the devices under test

☒ Yes
☐ No

The solar simulator spectrum was meticulously adjusted to maintain a spectral mismatch within 100 ± 3% for every 50 nm interval across the 300–1,200 nm range.

Explain why this information is not reported/not relevant.

6. Mask/aperture

Size of the mask/aperture used during testing

☒ Yes
☐ No

0.0515 and 1 cm² for solar cells, 17.7 and 81.5 cm² for solar modules.

Explain why this information is not reported/not relevant.

Variation of the measured short-circuit current density with the mask/aperture area

☐ Yes
☒ No

Report the difference in the short-circuit current density values measured with the mask and aperture area.

Negligible.

7. Performance certification

Identity of the independent certification laboratory that confirmed the photovoltaic performance

☒ Yes
☐ No

The champion solar cell device was certified by the Tianjin Institute Metrological Supervision and Testing, China, with an efficiency of 27.35%.

Explain why this information is not reported/not relevant.

A copy of any certificate(s)

☒ Yes
☐ No

Supplementary Figure 58.

Explain why this information is not reported/not relevant.

8. Statistics

Number of solar cells tested

☒ Yes
☐ No

15 devices for normal perovskite solar cells and 8 devices for solar modules.

Explain why this information is not reported/not relevant.

Statistical analysis of the device performance

☒ Yes
☐ No

Fig. 4b and Supplementary Figure 55.

Explain why this information is not reported/not relevant.

9. Long-term stability analysis

Type of analysis, bias conditions and environmental conditions

☒ Yes
☐ No

The operational stability was evaluated under continuous one-sun illumination (AM 1.5G, 100 mW cm⁻²) at maximum power point tracking (MPPT) and 50±5 °C, following the ISOS-L-1I protocol.

Explain why this information is not reported/not relevant.